

Artificial Intelligence

GLOSSARY 01 - SHEET 2 / 8

Machine-based systems that infer, from input, how to generate outputs that influence environments

PROFESSIONAL DEFINITION

An artificial intelligence system is a machine-based system that, for explicit or implicit objectives, infers from the input it receives how to generate outputs such as predictions, content, recommendations or decisions that can influence physical or virtual environments; different AI systems vary in their levels of autonomy and adaptiveness after deployment (OECD, 2024; Regulation (EU) 2024/1689, Article 3(1)).

WHO DEFINED IT ORIGINALLY

The definition in force for European business was drafted by the OECD's expert group for the 2019 Recommendation of the Council on Artificial Intelligence, revised in November 2023, and adopted with minor changes as the legal definition in Article 3(1) of the EU AI Act. The intellectual lineage runs to McCarthy, Minsky, Rochester and Shannon's (1955) Dartmouth proposal, which named the field.

WHY THIS DEFINITION WAS DEVELOPED

Regulators and standard-setters needed a definition that was technology-neutral and durable: narrow enough to exclude ordinary deterministic software, broad enough to survive the next architecture. Anchoring on inference - the system derives its outputs rather than having them enumerated by a programmer - achieved this, and the OECD-EU alignment gave firms one definition to satisfy across jurisdictions.

HOW IT IS USED

Apply the inference test to each candidate system to establish whether it falls inside the regulatory perimeter; systems that merely execute rules written by a human are outside it and need no conformity work.

WHEN IT IS USED

At procurement, at the start of any AI project intake, when compiling the AI system inventory required for governance, and whenever a vendor markets a product as artificial intelligence.

BY WHOM IT IS USED

Legal and compliance functions determining regulatory scope, procurement teams assessing vendor claims, and technology leaders maintaining the enterprise AI register.

FIGURE 1 - FROM INPUT TO ENVIRONMENTAL EFFECT

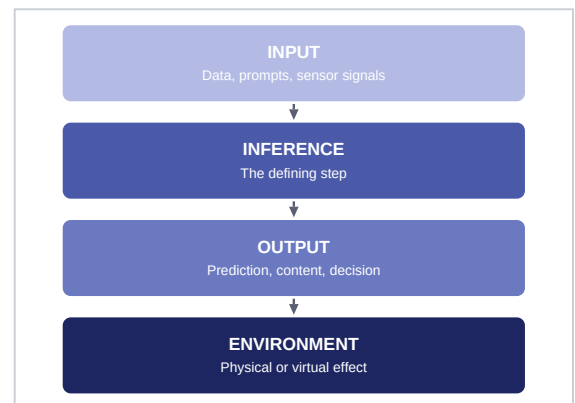


Figure 1. The inference test. A system is an AI system when the middle step is present: outputs are derived from input rather than enumerated in advance. Autonomy and adaptiveness vary along this chain and determine the obligations that attach to it (after OECD, 2024).

INNOVATION RELATIVE TO THE PREDECESSOR TOOL

The predecessor definitions were capability lists - systems that perform tasks normally requiring human intelligence - which shifted every time a task was automated, a problem long known as the AI effect. The OECD formulation replaces the moving human benchmark with a structural test based on inference and autonomy, so classification does not change as public expectations change. Its adoption into binding EU law gave the definition legal consequence rather than descriptive convenience.

RELATED DEFINITIONS IN THIS COURSE

[Digital Transformation - Unit 1](#)[AI Literacy - Unit 2](#)[General Purpose Technology - Unit 5](#)[EU AI Act - Unit 6](#)[Human-in-the-Loop - Unit 2](#)

SOURCES (HARVARD REFERENCING STYLE)

European Parliament and Council (2024) Regulation (EU) 2024/1689 laying down harmonised rules on artificial intelligence. Official Journal of the European Union, L, 12 July 2024.

McCarthy, J., Minsky, M.L., Rochester, N. and Shannon, C.E. (1955) A Proposal for the Dartmouth Summer Research Project on Artificial Intelligence. Hanover, NH: Dartmouth College.

OECD (2024) Explanatory Memorandum on the Updated OECD Definition of an AI System. OECD Artificial Intelligence Papers No. 8. Paris: OECD Publishing.

Russell, S. and Norvig, P. (2021) Artificial Intelligence: A Modern Approach. 4th edn. Harlow: Pearson.